

Part 1: SOFTWARE PROJECT PLANNING

Project Description

The Campus Event Tracker is a user-friendly web application that allows students and campus organizations to add, view, and manage upcoming events. Users can input event titles and dates, and the application displays all events in a dynamic, interactive list. Events are stored locally to ensure persistence across browser sessions.

Project Scope – Functional Requirements

1. Users can add new events with a title and date.
2. The application displays a dynamic list of all saved events.
3. Events are stored in local storage to persist across sessions.

Task Breakdown & Estimation (Planning Poker)

Task	Description	Estimation (Story Points)
Create EventItem Model	Define the event structure with <code>id</code> , <code>title</code> , and <code>date</code> .	2
Build Event Tracker Component	Create a component for adding and displaying events.	5
UI Layout & Styling	Design and apply CSS/Bootstrap for a modern interface.	3
Implement Event Logic	Add TypeScript methods to store and display events dynamically.	5
Test & Debug	Test input validation, event list, and storage functionality.	3

Total Story Points: 18

Part 2: SOFTWARE ARCHITECTURE DESIGN

Chosen Architecture Style

Component-Based Architecture

Explanation

Angular employs a component-based architecture, where the UI is divided into modular, self-contained components. Each component contains its own HTML template, CSS styling, and TypeScript logic. Components can communicate via inputs and outputs, allowing for modular, reusable, and maintainable code.

Justification

This architecture is ideal for a small application like the Campus Event Tracker because it allows each feature—such as the event tracker—to be developed and tested independently. Component-based design ensures separation of concerns, promotes reusability, and makes the application easy to maintain and expand in the future.

End of Project Plan